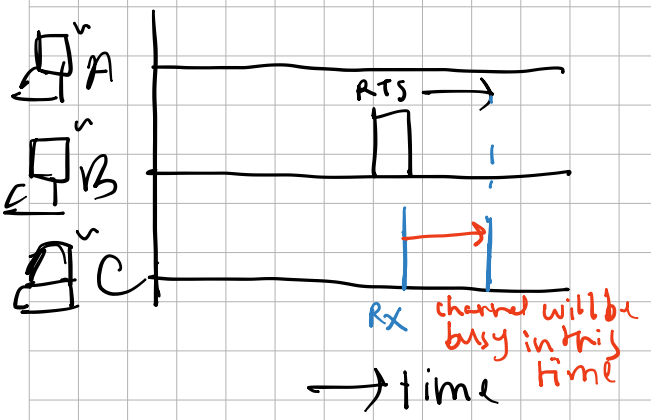
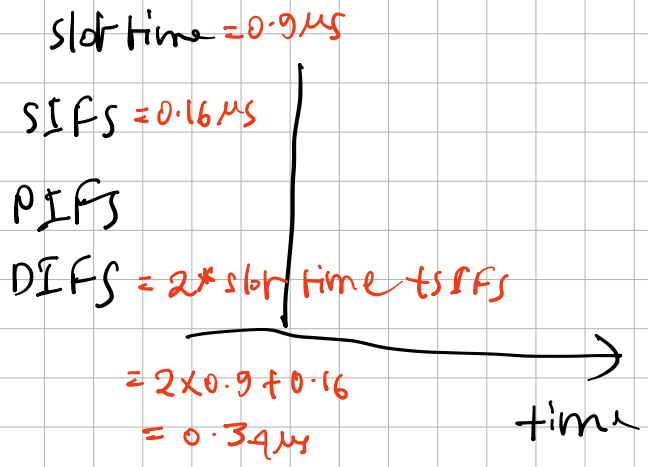
19th June, 25:





IEEE 802.11 → standard for WLAN

↳ affects lower layers ⇒ physical and MAC sublayer

↳ due to a change in context (wired to wireless)

Arch < BSS, IBSS

CF < DCF, PCF

BSS → Basic service set (AP exists)

IBSS → Independent Basic service set (AP doesn't exist)

- we need to follow a set of rules so that conflict can be reduced

so we come up with coordination functions

↳ DCF – QoS x (general services)

↳ PCF – QoS ✓

- CF is maintained in the MAC sublayer.

In order to reduce collision, we listen before sending, via CCA (clear channel assessment)

↓

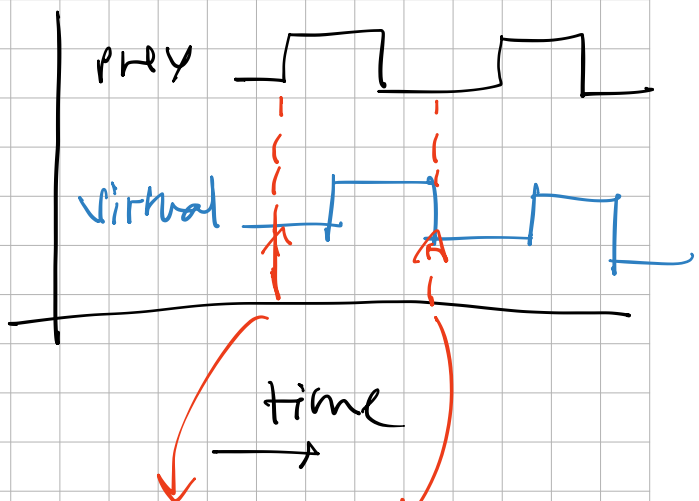
voltage level is measured

If voltage level > threshold ⇒ busy
< ⇒ non busy

determined by network administrator

- a few different types of signals might exist
  - microwave signal
  - homogenous signals (other signals)
  - NAV signals

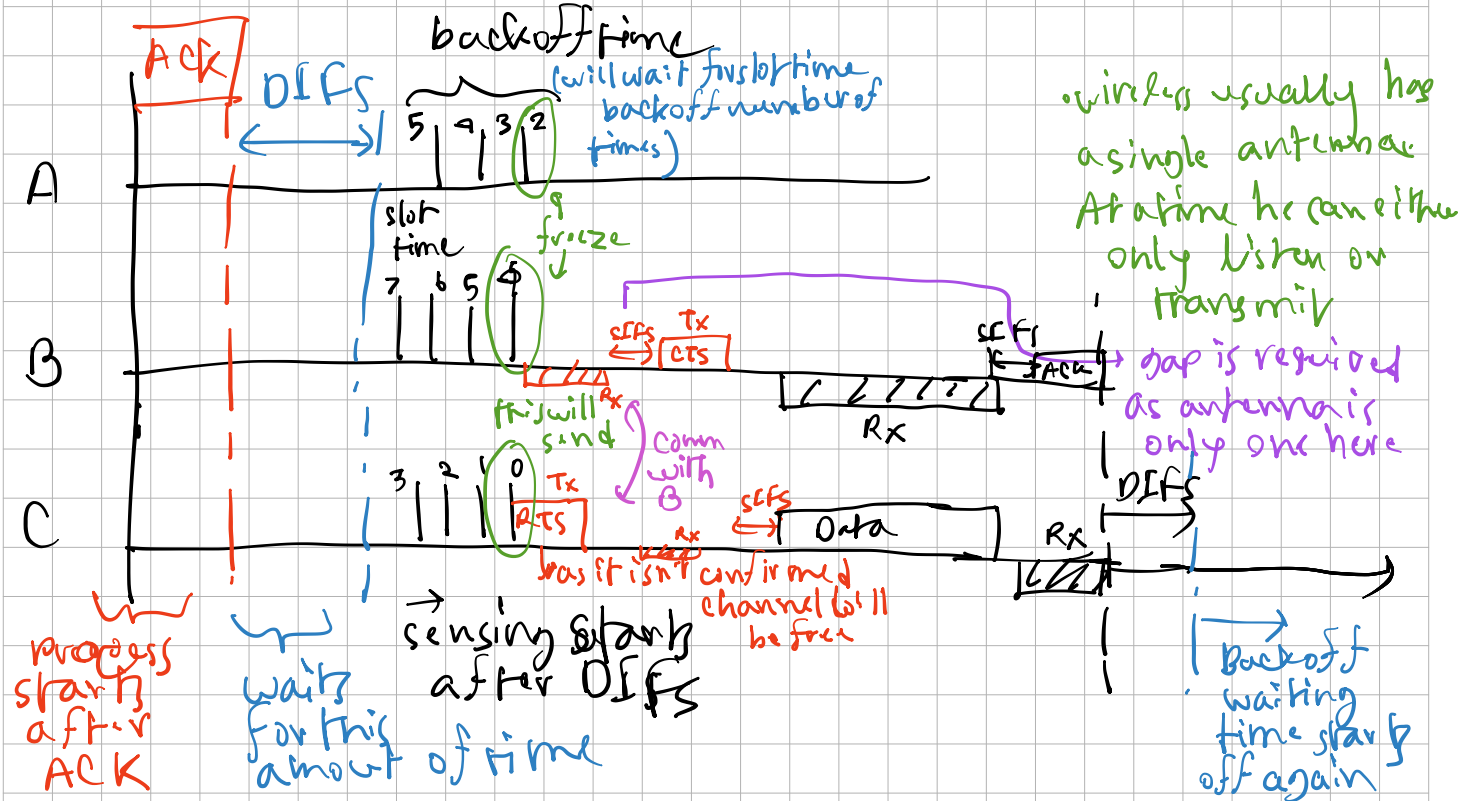
- CCA
  - physical
    - noise detection
    - preamble detection
  - virtual
    - NAV



- ACK isn't necessary in wired, as success/failure can be directly observed

  In wireless, ACK is explicitly needed.
- Similarly, RTS/CTS isn't necessary in wired, but we send it to avoid hidden terminal problem

Timing and Interframe space:



- wireless usually has a single antenna. At a time he can either only listen or transmit
- gap is required as antenna is only one here

contention window $CW = CW_{min-1}$

amount of time channels will be waiting $r = \min(2^{n-1} CW, CW_{max})$ (n = number of attempt, 1024)

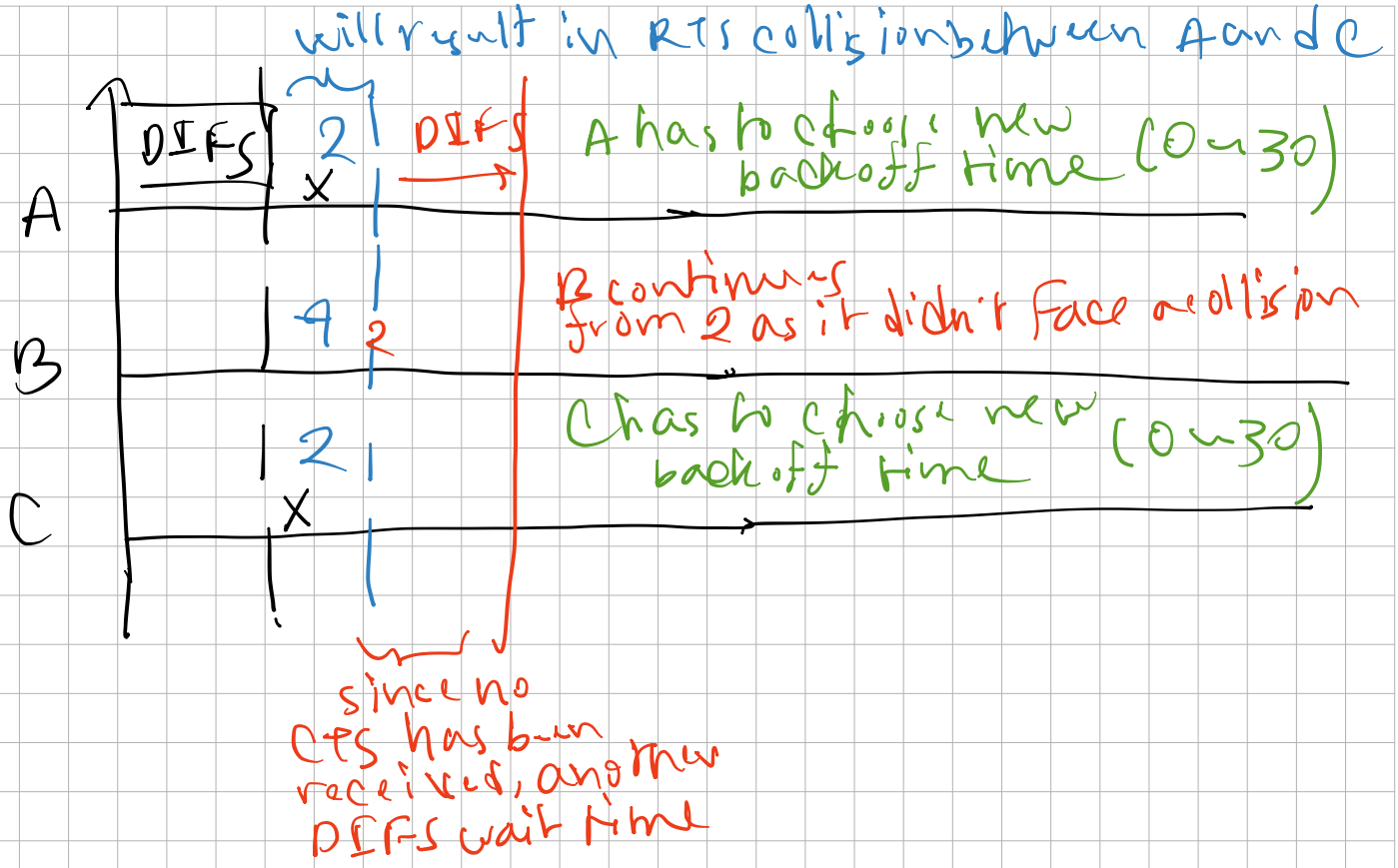
$= \min(2^{1-1} \cdot 15, 1024)$

for first attempt

$\therefore$ range $r = 15$

$\therefore$ backoff count value $= \text{rand}(0 \sim 15)$

waiting time count



- there is a high channel wastage in this method due to backoff time. (as there is no traffic controller)

15th July, 2025:

Current Wifi channel Contention:

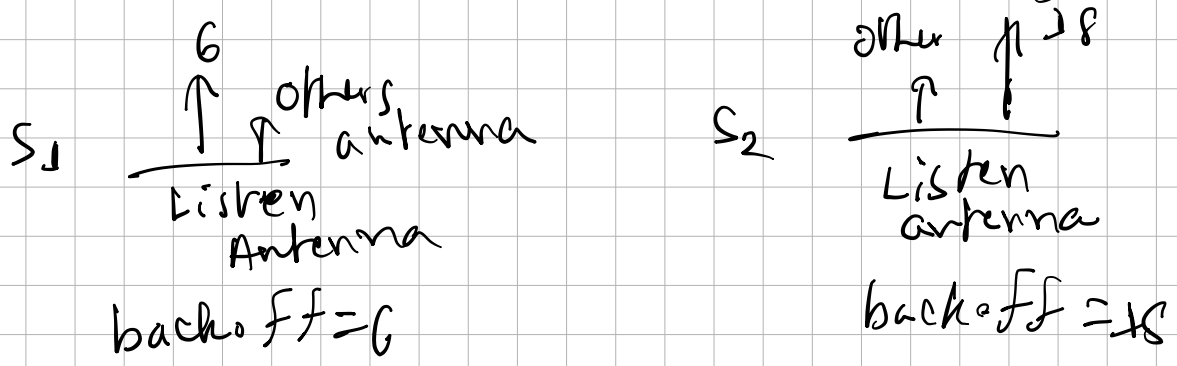
problem: multiple pcds transmitting at the same time

- how to resolve backoff in frequency domain?

Back2F:

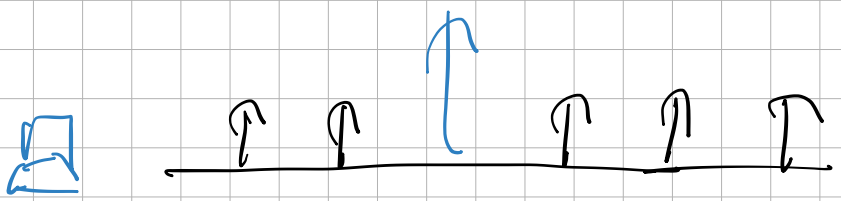
In time domain backoff → backoff time waiting was necessary

frequency domain backoff → chooses random number of subcarrier to transmit their signal.



∴ S1 is the winner

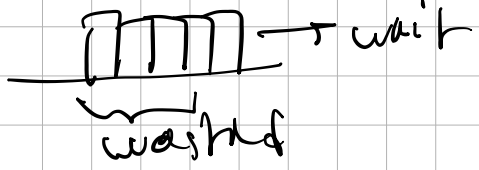
Example:



Rank in TDMA: 3

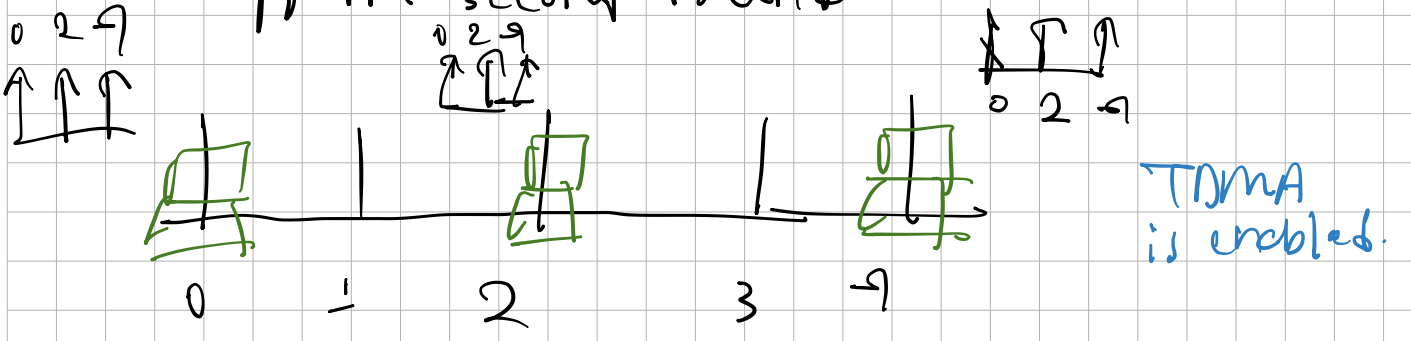
In TDMA every station knows its schedule and transmits only at that time.
This schedule is repeated once all stations have finished transmitting.

- In time domain multiple slot times were necessary to solve the contentions.



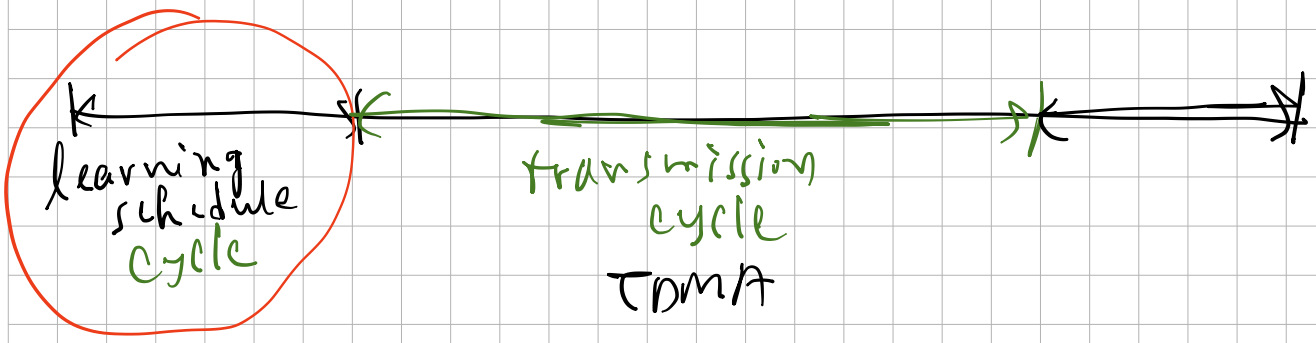
In frequency domain, contention is solved in only one slot.

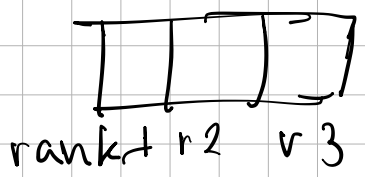
- We can also introduce a second round of contention.
  - some winners and a few stations are allocated to the second round.



- Once stations have been selected all subcarriers can be used.
- high frequency subcarriers are easily affected in wireless environments.

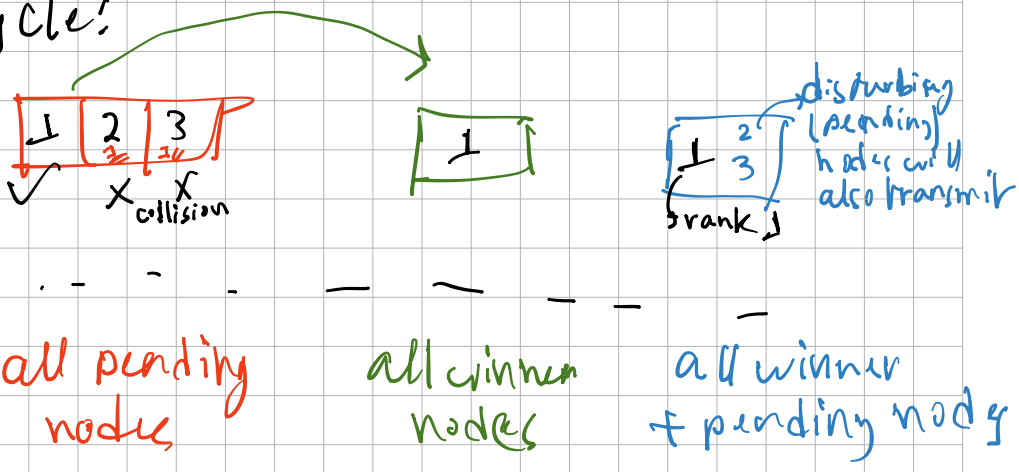
17th July, 25:





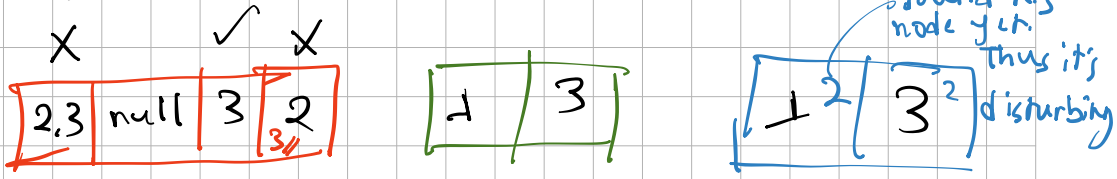
in a learning cycle!

learn cycle



all cycles are completed horizontally. meaning, all phases of a cycle will complete before another cycle can begin.

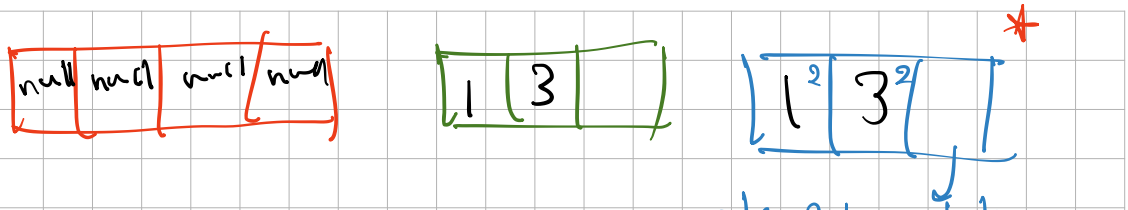
∴ cycle 2



∴ winner transmits on his dedicated slot and pending transmits every where

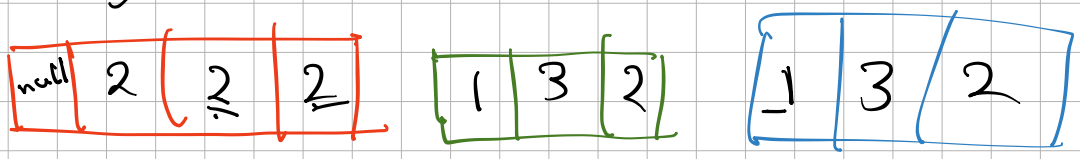
∴ Once we see that there aren't any disturbing nodes, we'll know that the transmission has finished

cycle 3



node 2 doesn't know that it's alone. Thus, there is no one to disturb here.

cycle 4



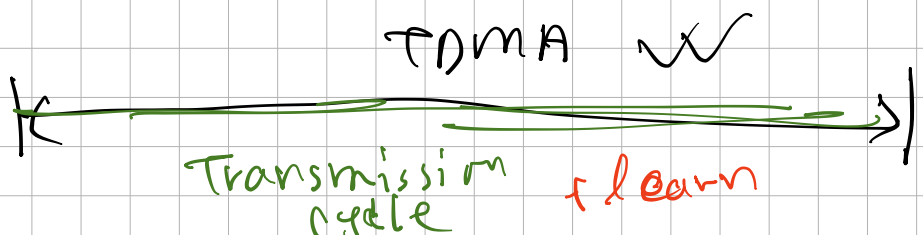
Thus, the schedule has been learned. This is the schedule that will be used during transmission.

∴ This TDMA solution eliminates OFDM problems.
However, learning phase is very complex and strict time synchronization is necessary

Also not efficient in dynamic environments
→ Every time we have a new node entering the network, this learning cycle will take place.

∴ this is good for stable networks.

- Another approach:

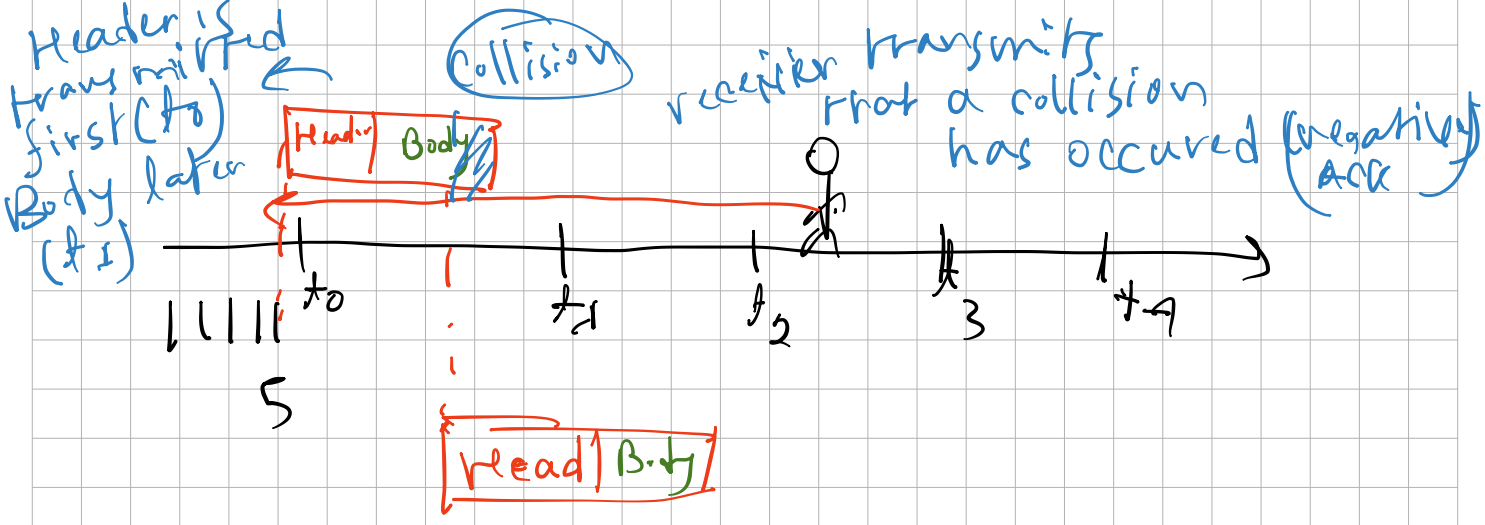


∴ Transmission and learning takes place at the same time.

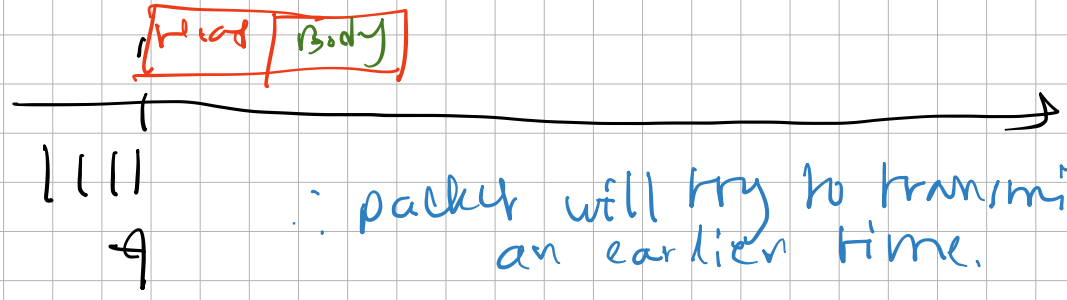
· EMAC:

- If the front part is intact in a collision, the receiver will be able to send a negative acknowledgement to the transmitter.

- Transmitter will learn this information and transmit a bit earlier next time.



∴ next transmission:



∴ packet will try to transmit at an earlier time.

∴ this is transmission and learning.

This doesn't require a complex learning and requires more than one slot time
∴ slow process.

Will not work if header part is collided.
But works for dynamic network.

24th July 25:

- We don't want:
  - OFDM (specific physical layer characteristic)
  - Additional HW (Antenna)
  - Explicit and complex learning cycle
  - Complex deduction
  - Time sync (strict)

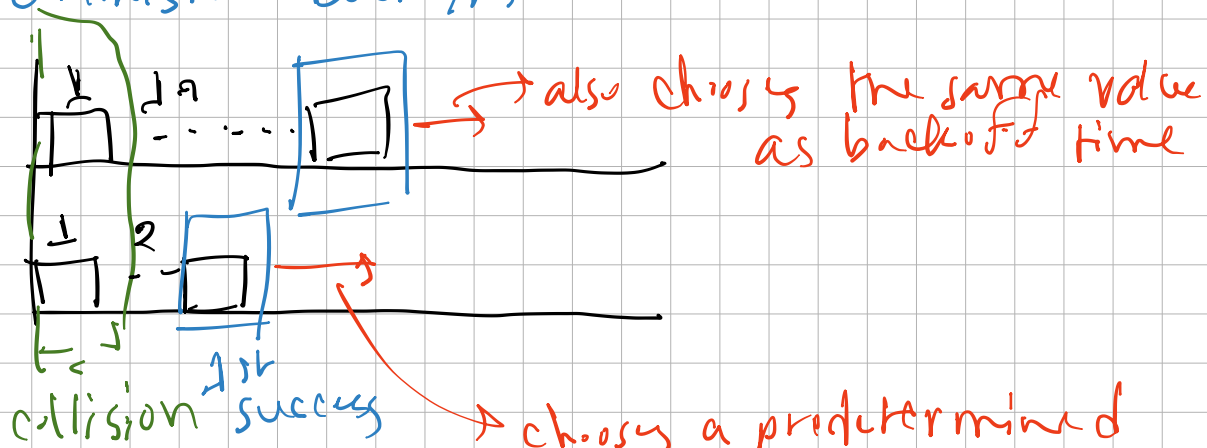
- We want:
  - TDMA scheduling
  - No collision
  - less idle slot (less channel time wastage)

- We want CSMA as it's very simple, however can't guarantee no collisions will occur

  Thus it's not efficient.

  However CSMA/CA already satisfies all this

Deterministic Backoff:



∴ all consecutive transmissions will happen in their own designated time

∴ no further collision will occur.

if a new station joins, it'll have collisions at the start and eventually join the safe zone.

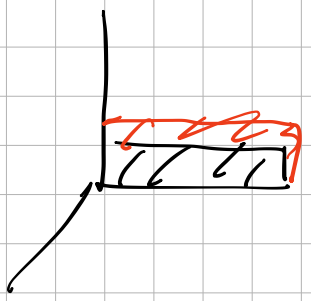
- if initial deterministic value is 7 and they reach a safe zone once, but then reach an unstable state again after some time it means that new stations have joined the station list.

  ∴ thus backoff time will increase.

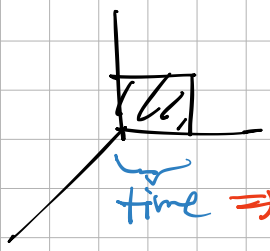
- In order to decrease slot time, we need AP's help (Access point = beacon)

29th July, 25:

Non orthogonal multiple access (NOMA):



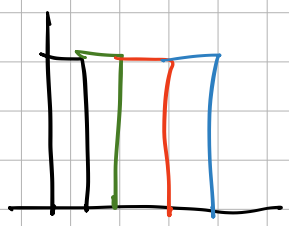
→ overlapping orthogonality



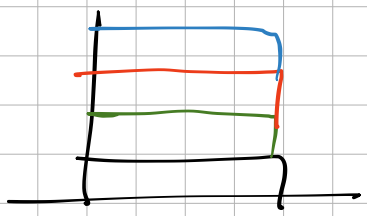
⇒ smallest unit of time that is unshavable

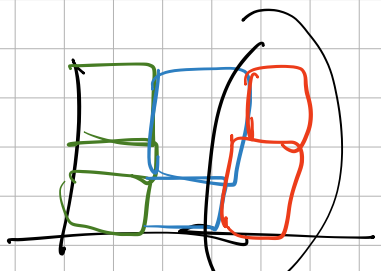
NOMA:
- Power level NOMA
- Code level NOMA

∴ different people use different power level in smallest unity of time.



OFDMA





NOMA

multiple signals at a time

∴ smallest non shavable unit can also be shared among multiple users.

∴ NOMA is like complex CDMA (code domain)

7 Aug '25:

Introduction of VANET:

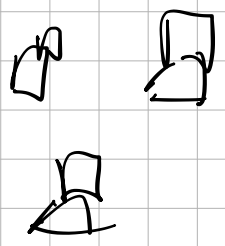
- Smart city networks are ad-hoc.
- Difference between VANET and MANET

VANET



- speed of movement is very high
- movement is predictable
- all vehicles need to communicate with each other and be failsafe. Communication needs to be perfect and fast.

MANET



- speed is low
- unpredictable movement
- no life threat here
- impact of mobility is low

∴ That's why we need a seperate network for vehicles which is seperate from the usual Ad hoc network.

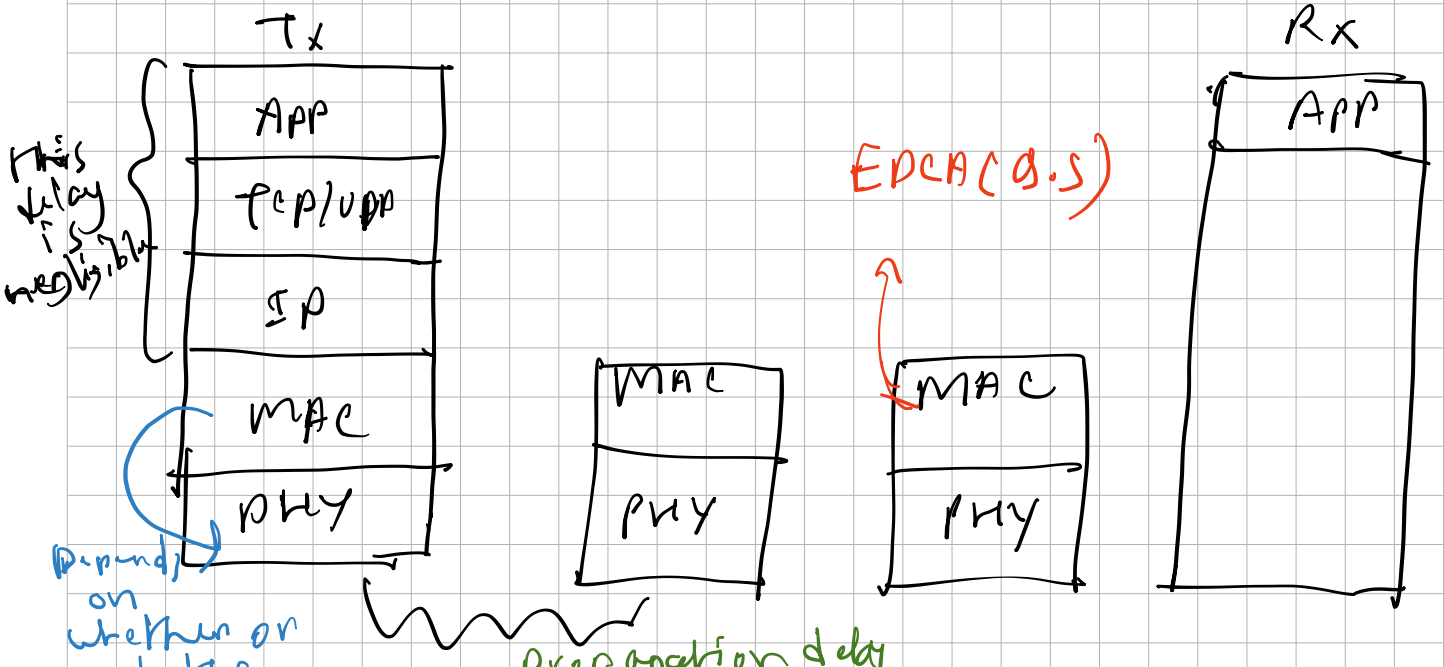
- Communication standards were discussed in lab:
  - wifi solutions: IEEE 802.11p
    - DSRC / WAVE (need to know full form)
      - These are the wifi based communication standard
        - USA
        - Europe → C-ITS (comm for intelligent transport system)

- Cellular solutions:
  3GPP (3rd Generation partnership project, 3G, 4G are derived from this)

  PC5 , UU } These are the two modes of operation which supports direct communication

Both LTE and 4G have these two modes (PC5, UU)

- Delay: Wifi ⇒ 10ms latency
  Cellular ⇒ 5-10ms latency

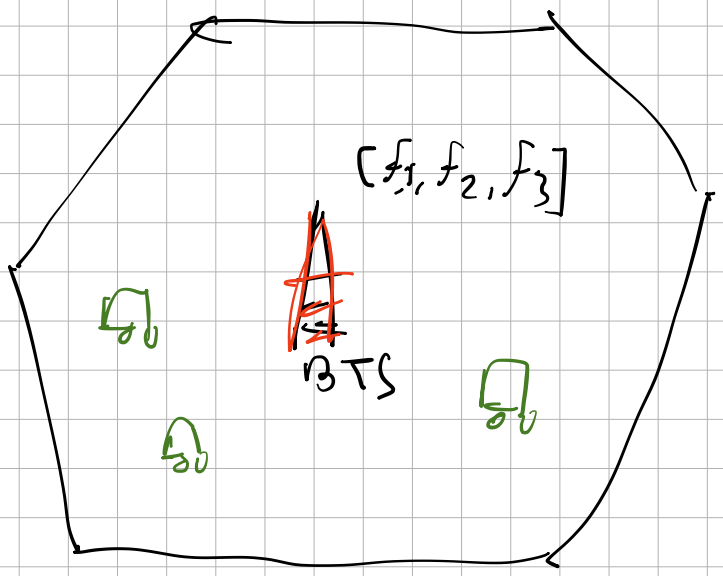


⇒ The delay we experience here is variable

∴ MAC layer ⇒ variable delay
everywhere else ⇒ constant delay

Under ~~DCA~~ mac layer is distributed, thus the delay is moderately high. (Random Access) CSMA

- For cellular networks:



This channel allocation is like resource allocation.

Everything is determined by BTS, which means it's centralized

∴ Cellular has centralized resource sharing / allocation, that's why there is low delay.
This is FDMA.

↓

If there are more vehicles, we can use TDMA on top of this as well.

On top of this, we can also use CDMA.

↓

OFDM, MIMO can be used on top of this.

Thus, this resource assignment is entirely central.

∴ wifi → CSMA
Cellular → CDMA, TDMA, FDMA etc.

- VAMEF has relatively lower performance but it's good enough to be safety critical.

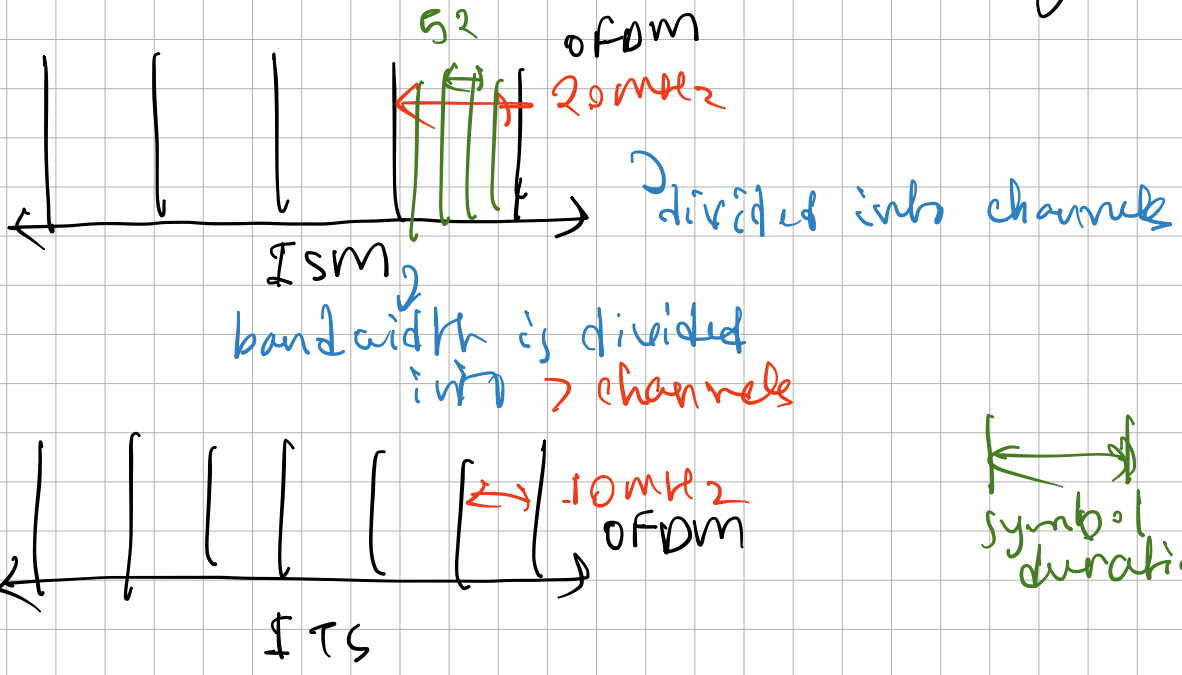
We also don't need to pay for this.

14th August, 25;

- IEEE 802.11p standard works in the lower two layers.

  802.11p MAC ← → 802.11p PHY

- WSMP is the alternative for TCP, UDP, IPv6
- Application layer has some security issues.
- CAMEP – no need to connect to AP / BSS
  – no need for a lot of handshaking
-



- Advantages and disadvantages of cellular and wifi standard
- DSRC → Channel switching timing.
  SCH → Service channel.